

Death by design: where curcumin sensitizes drug-resistant tumours.



Chemotherapy remains the core of anticancer treatment. However, despite the tremendous strides made in the development of targeted anticancer therapies, emergence of resistance to chemotherapeutic drugs is still a major obstacle in the successful management of resistant tumours. Therefore, profound investigation into the in-depth molecular mechanisms of drug resistance is essential and may hopefully translate into effective therapies that can flip the switch from drug resistance to susceptibility. Mechanistically, resistance phenomena may be explained by (i) overexpression of drug efflux pumps, (ii) enhanced drug detoxification, (iii) rapid DNA repair efficiency, (iv) defects in apoptosis regulation, and (v) active cell survival signals. Several adverse effects associated with multidrug resistance and the need for safe multi-targeted anticancer drugs instigated the use of the phytochemical, curcumin, the yellow pigment of the spice turmeric, which has pleiotropic activities. We performed a structured literature review using PubMed and Medline searches with secondary review of cited publications, identifying studies on the role of curcumin in conquering drug resistance in cancer. This review describes how curcumin sensitizes cancer cells through regulation of multiple multidrug resistance pathways, thus employing one drug for multiple targets. Curcumin helps the cancer cells to regain their 'forgotten' apoptosis, modulates drug-target interaction at different levels, restrains survival pathways when their proteins are overexpressed, and finds an alternate way to carry forward the process of sensitization of different resistant tumours. Additionally, the review dissects the role of curcumin, if any, in targeting the major culprit of drug resistance, cancer stem cells (CSC), thereby circumventing resistance. Taken together, this review strongly suggests that curcumin is a promising chemosensitizing agent and that the unique properties of curcumin may be exploited for successful management of resistant tumours.